

Structural Failure Diagnosis and Structure-Aware Densification for 3D Gaussian Splatting under Real Blur

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Abstract— Blur-aware 3D Gaussian Splatting (3DGS) rests on a principle dating to Lee and Lee (CVPR 2013): motion blur is a geometric consequence of camera motion and scene depth, and should be modelled inside the reconstruction objective. On RealX3D, a recent benchmark of physically captured degradation, this does not reliably pay off—rankings invert between motion and defocus blur, and under mild blur plain 3DGS attains the highest PSNR of the six methods evaluated. RealX3D scores these methods with image-space metrics only, so it is unknown whether geometry fails in a structure-dependent way that scene-averaged PSNR conceals. We propose a two-stage project: a stratified geometric evaluation partitioning RealX3D's laser-scanned ground truth by local curvature and surface proximity, reporting accuracy, completeness and F-score per stratum; and, guided by that diagnosis, a structure-aware densification rule that reallocates Gaussian primitives toward geometrically fragile regions under blur.

Keywords— 3D Gaussian Splatting, motion blur, defocus blur, geometric evaluation, adaptive density control, robust 3D reconstruction

I. INTRODUCTION AND RELATED WORK

Blur is unavoidable in handheld and robot-mounted capture, and corrupts reconstruction by destroying the local structure that correspondence and photometric optimisation depend on. Lee and Lee [1] observed that blur is not an independent 2D corruption but a geometric one: given camera motion and exposure time, a pixel's blur kernel is the projected trajectory of its 3D scene point during exposure, determined in closed form by that pixel's depth. Estimating blur becomes estimating depth.

That principle—parameterise blur by scene geometry rather than as a free 2D kernel—underpins the modern lineage. Deblur-NeRF [4] and its successors introduced learned per-ray kernels and bundle-adjusted exposure trajectories; on the 3DGS [2] side, Deblurring-3DGS [5], BAD-Gaussians [6], BAGS [7] and CoCoGaussian [8] model blur respectively through per-Gaussian covariance offsets, SE(3) exposure trajectories, per-pixel multi-scale kernels, and the depth-dependent circle of confusion; see [11] for a survey.

RealX3D [3] is the first real-capture benchmark for physically produced degradation, using a programmable rail to re-run identical trajectories and obtain pixel-aligned degraded and reference pairs, plus laser-scanned ground truth. Its blur results are inconsistent in an informative way. Under strong motion blur (train-view PSNR) Deblurring-3DGS leads at 20.64 dB while BAGS (18.75) falls *below* vanilla 3DGS (20.33); under strong defocus the order reverses, with BAGS leading (21.01) and Deblurring-3DGS dropping beneath vanilla 3DGS (20.83). Under both *mild* settings plain 3DGS attains the highest PSNR of all six methods. No method dominates, and the benchmark reports this without explaining it. The NTIRE 2026 3DRR Challenge [13] built on RealX3D but ran only low-light and smoke tracks, leaving blur comparatively unexamined.

Evaluating reconstruction on structurally challenging regions rather than only in aggregate is established practice, which we adopt rather than claim: RNb-NeuS [9] reports Chamfer and normal error restricted to high-curvature and low-visibility areas, and geometric assessment of aerial NeRF [10] uses accuracy and completeness beyond image-space metrics. Neither has been applied to blur-degraded 3DGS, and RealX3D reports no geometric metrics at all for the six methods above—its accuracy and F-score figures cover only feed-forward models. That gap motivates both stages of this project.

II. PROBLEM AND PROPOSED WORK

A. Problem Formulation

Scene-level image metrics may underrepresent failures on thin or geometrically complex structures, since such regions occupy few image pixels while being the first to lose high-frequency support under blur; a 0.3 dB difference in scene-averaged PSNR could then conceal opposite behaviour on the structures that matter most downstream. We ask: (Q1) does the ranking of blur-aware 3DGS methods change when error is scored per structural stratum rather than per scene; and (Q2) can a targeted change to primitive allocation recover the structures degradation destroys?

B. Stage 1: Stratified Geometric Evaluation

We partition each scene's ground-truth geometry along independent structural axes. The *core axes* are *local curvature*, estimated by a robust neighbourhood-based surface-variation measure with strata validated for stability across neighbourhood scales, and *surface proximity*, the distance to the nearest independent surface, separating isolated geometry from tightly spaced structure. A third axis, *local thickness* via the Shape Diameter Function, is included only if the released mesh passes watertightness and manifoldness checks; because it is frustum-trimmed it may not, so we treat it as conditional rather than core.

Each axis is quantile-binned into three strata and analysed *marginally*—separate analyses rather than a joint 27-cell partition, which would leave cells sparsely populated across eight scenes. Every reconstruction is converted to a point cloud by one common procedure, rendering depth from training viewpoints and fusing back-projected points, rather than reading Gaussian centres, since

centre clouds reflect primitive placement rather than reconstructed surface. Clouds are ICP-aligned to the laser scan per RealX3D's protocol before per-stratum metrics are computed.

C. Stage 2: Structure-Aware Densification

Vanilla 3DGS clones or splits a Gaussian when its accumulated view-space positional gradient exceeds a fixed threshold. Blur attenuates high-frequency content, so gradients are suppressed precisely where geometric detail is scarcest: thin and high-curvature regions may never densify, while large textured surfaces keep accumulating primitives. We will implement a modified rule that (i) normalises the densification signal by a per-view sharpness estimate, so heavily blurred views do not dilute the gradient statistic, and (ii) lowers the effective threshold for candidates whose neighbourhood indicates structurally fragile geometry. The intervention is deliberately small—a decision rule inside an existing 3DGS implementation, no new network—and is benchmarked against MCMC-based densification, which replaces the clone/split heuristic altogether within a broader deblurring pipeline [12]. Stage 2 is *guided by* rather than dependent on Stage 1: if stratification shows thin structures are *not* disproportionately degraded, the same instrumentation tests the converse allocation hypothesis.

D. Supporting Diagnostic Hypotheses

H1: adaptive density control (ADC) contributes. The baselines interact with densification differently—Deblurring-3DGS [5] adds points and prunes by depth, BAD-Gaussians [6] rescales the threshold by virtual-view count, CoCoGaussian [8] spawns multiple Gaussians per primitive, BAGS [7] leaves ADC untouched—but they also differ in blur modelling and optimisation, so cross-method differences cannot be attributed to ADC alone; we test whether Gaussian population *correlates* with per-stratum error, establishing causality only by ablation within a single method. *H2: blur-model mismatch*—RealX3D integrates exposure over a known rail displacement using 64 sub-poses and its defocus is global, so recovered kernels can be checked against known parameters. *H3: trajectory drift*—since [3] supplies reference-view-derived poses, we measure absolute trajectory error of trajectory-optimising methods against that initialisation.

E. Scope, Risk and Fallbacks

Core deliverables are the geometry harness, the Stage 1 evaluation, and the Stage 2 densification modification with a controlled comparison. The guaranteed grid is four methods—vanilla 3DGS, the strongest motion and defocus baselines from [3], and ours—with the full six-method grid an extension; H1 is tested within Stage 2, while H2 and H3 are reach items. The dominant unbounded risk is building heterogeneous CUDA stacks whose 2024-era kernels may not compile for recent GPUs; we target older cluster nodes and set a week-3 checkpoint at which non-building baselines are dropped. Mesh topology risk is handled by the core/optional axis split above.

III. PROPOSED EVALUATION

Data. The RealX3D blur subset: 8 scenes $\times$ {motion, defocus} $\times$ {mild, strong} at quarter resolution ($\approx 1800 \times 1200$), with ~ 23 –31 training and 4–6 test frames per scene, COLMAP poses, laser-scanned mesh and 16-bit metric depth. The reduced grid is 4 methods $\times$ 8 scenes $\times$ 4 conditions = 128 runs, with the undegraded reference capture as control.

Metrics. Per-stratum accuracy, completeness and F-score. We adopt RealX3D's 5 cm threshold and also report F1 at 2 cm and 10 cm, since a single threshold is sensitive to scene scale. Scene-averaged PSNR/SSIM/LPIPS are retained for comparability with [3]; Gaussian population statistics support H1.

Reporting. Per-scene results accompany aggregate means, with paired per-scene differences between methods, so conclusions are not driven by one or two scenes. Before any stratified number is trusted we reproduce RealX3D's photometric table to within ≈ 0.3 dB and cross-check the harness against reference implementations.

Success criteria. (i) Reproduction within tolerance; (ii) a per-stratum comparison establishing whether ranking is stratum-dependent; (iii) a measurable per-stratum improvement from the Stage 2 modification, or a documented account of why it fails. A negative Stage 1 result—rankings stable under stratification—remains reportable, since it would establish that scene-averaged metrics are adequate here, which is currently assumed rather than demonstrated.

REFERENCES

- [1] H. S. Lee and K. M. Lee, "Dense 3D reconstruction from severely blurred images using a single moving camera," in *Proc. CVPR*, 2013, pp. 273–280.
- [2] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, "3D Gaussian splatting for real-time radiance field rendering," *ACM Trans. Graph.*, vol. 42, no. 4, 2023.
- [3] S. Liu *et al.*, "RealX3D: A physically-degraded 3D benchmark for multi-view visual restoration and reconstruction," arXiv:2512.23437; *IJCV*, 2026.
- [4] L. Ma *et al.*, "Deblur-NeRF: Neural radiance fields from blurry images," in *Proc. CVPR*, 2022.
- [5] B. Lee, H. Lee, X. Sun, U. Ali, and E. Park, "Deblurring 3D Gaussian splatting," in *Proc. ECCV*, 2024; arXiv:2401.00834.
- [6] L. Zhao, P. Wang, and P. Liu, "BAD-Gaussians: Bundle adjusted deblur Gaussian splatting," in *Proc. ECCV*, 2024; arXiv:2403.11831.
- [7] C. Peng *et al.*, "BAGS: Blur agnostic Gaussian splatting through multi-scale kernel modeling," in *Proc. ECCV*, 2024; arXiv:2403.04926.
- [8] J. Lee *et al.*, "CoCoGaussian: Leveraging circle of confusion for Gaussian splatting from defocused images," in *Proc. CVPR*, 2025; arXiv:2412.16028.
- [9] B. Brument *et al.*, "RNb-NeuS: Reflectance and normal-based multi-view 3D reconstruction," in *Proc. CVPR*, 2024; arXiv:2312.01215.
- [10] Y. Xu *et al.*, "Multi-tiling NeRF: geometric assessment on large-scale aerial datasets," *The Photogrammetric Record*, 2024.
- [11] W. Kwon *et al.*, "R3eVision: A survey on robust rendering, restoration and enhancement for 3D low-level vision," arXiv:2506.16262, 2025.
- [12] G. R. Matta, T. Reddypalli, V. D. Madhuri, and K. Mitra, "GeMS: Efficient Gaussian splatting for extreme motion blur," arXiv:2508.14682, 2025.
- [13] S. Liu *et al.*, "NTIRE 2026 3D restoration and reconstruction challenge: RealX3D results," in *Proc. CVPR Workshops*, 2026; arXiv:2604.04135.